PROJECT MANAGEMENT AND PLANNING

COMP6100001 - SOFTWARE ENGINEERING
SESSION 11

UNIVERSITAS **BINA NUSANTARA**

SUBJECT MATTER EXPERT

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SESSION 11 – PROJECT MANAGEMENT AND PLANNING

LEARNING OBJECTIVES

At the end of this session, students will be expected to:

- Understand the way of thinking in software engineering
- Understand the fundamental software project management
- Know how to be a good leader and lead the team in software project

SESSION 11 – PROJECT MANAGEMENT AND PLANNING

CONTENT OUTLINE

- Introduction to Project Management Concepts
- Importance of Project Planning in Software Engineering
- Roles & responsibilities
- Work Breakdown Structure (WBS)

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

DEFINITION AND PRINCIPLES OF PROJECT MANAGEMENT

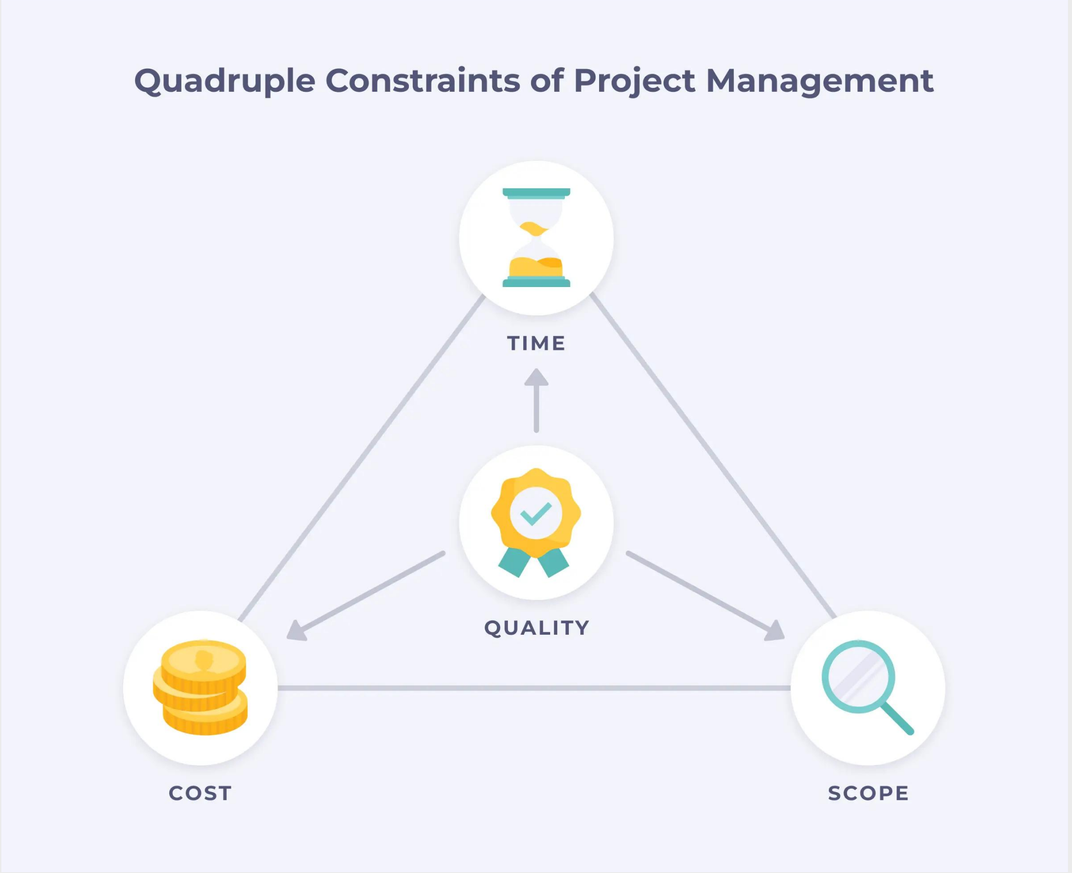
- Definition:
  - Project Management is the application of knowledge, skills, tools, and techniques to project activities to meet the project requirements.

- Key Principles:
  - Project Initiation: Defining the project at a broad level.
  - Project Planning: Establishing the scope, objectives, and procedures.
  - Project Execution: Coordinating people and resources.
  - Project Monitoring and Control: Tracking progress and performance.
  - Project Closure: Formalizing acceptance and bringing the project to an orderly end.

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PROJECT CONSTRAINTS

- Scope: Defines what is included in the project.
- Time: The schedule for completion.
- Cost: Budget allocated.
- Quality: The standards and criteria to meet.



INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

THE MANAGEMENT SPECTRUM

- Software project management is an essential part of software engineering
- Effective software project management focuses on the four P's:
  - People
  - Product
  - Process
  - Project

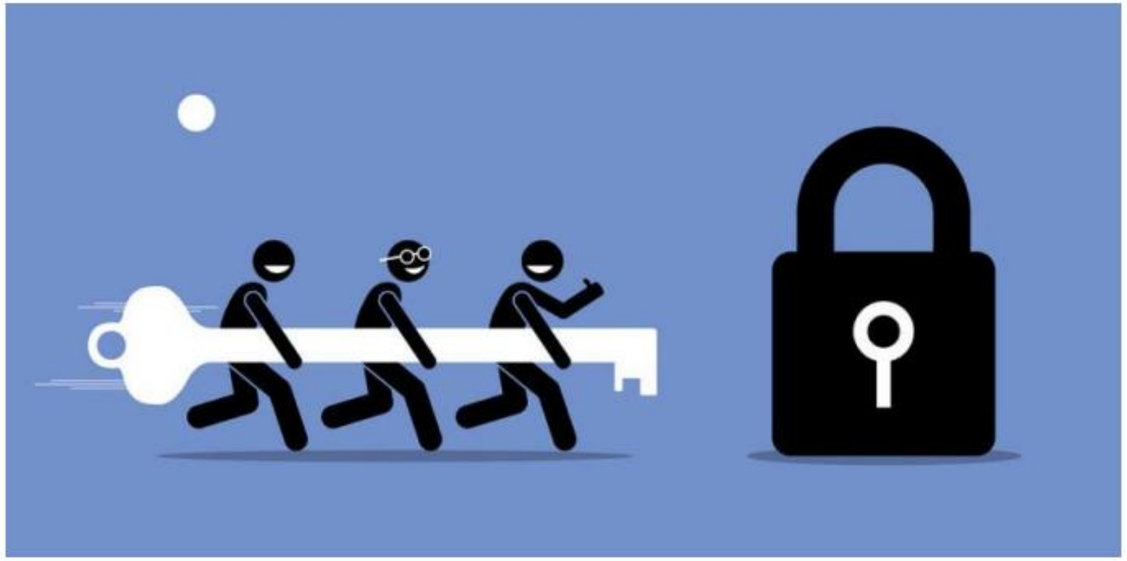
INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

THE FOUR P'S: PEOPLE

- The Stakeholders → participate in the software process and the way they are organized to perform effective software engineering:
  - Senior managers (product owners) who define the business issues that often have a significant influence on the project
  - Project (technical) managers (Scrum masters or team leads) who must plan, motivate, organize, and coordinate the practitioners who do software work.
  - Practitioners who deliver the technical skills that are necessary to engineer a product or application.
  - Customers who specify the requirements for the software to be engineered and other stakeholders who have a peripheral interest in the outcome.
  - End users who interact with the software once it is released for production use.

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PEOPLE: WORKING SOLO VS AS A TEAM



INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PEOPLE: WORKING AS A TEAM

- Design & Implement Software
  - Establish a collaboration process
  - Meet with the team
  - Choose a leader
  - Divide work and integrate
  - Share knowledge
  - Resolve conflicts

PEOPLE: HOW TO LEAD A TEAM

Managers and Tech Leads (and Both)

- Experienced manager comes in to run a team
- An individual contributor is promoted into a leadership position (usually of a smaller team)
- Both roles can be filled by the same person: a Tech Lead Manager (TLM)
- On larger teams,
- An experienced people manager will step in to take on the management role
- A senior engineer with extensive experience will step into the tech lead role

THE ENGINEERING MANAGER

- Software engineering managers should have an engineering background
- An engineering manager is responsible for the **performance**, **productivity**, and **happiness** of every person on their team—including their tech lead—while still making sure that the **needs of the business** are met by the product for which they are responsible

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

THE TECH LEAD

- The tech lead (TL) of a team → will often report to the manager of that team
- Responsible for the **technical aspects** of the product, including **technology** decisions and choices, **architecture**, **priorities**, **velocity**, and **general project management**
- Most TLs are also individual contributors, which are often forces to choose between doing something quickly themselves or delegating it to a team member to do (sometimes) more slowly
- Delegation is most often the correct decision for the TL as they grow the size and capability of their team

THE TECH LEAD MANAGER

- Tech Lead Manager (TLM) → a single person who can handle both the people and technical needs of their team
- TLM must know how to balance individual work, delegation, and people management

ANTIPATTERN

- Don’t follow these to be a successful manager:
  - Hire Pushovers
  - Ignore Low Performers
  - Be Everyone’s Friend
  - Compromise the Hiring Bar
  - Treat Your Team Like Children
- Why shouldn’t we do those patterns? Unleash your opinions!

POSITIVE PATTERN

- Follow these to be a successful manager:
  - Lose the Ego
  - Be a Zen Master
  - Be a Catalyst
  - Remove Roadblocks
  - Be a Teacher and a Mentor
  - Set Clear Goals
  - Be Honest
  - Track Happiness

- How are those patterns can help us to be a successful manager?

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

AS A LEADER

- Delegate, but get your hands dirty
  - Let the team know how skillful and dedicated (and humble) you are like jumping in and actually doing some hard work
- Seek to replace yourself
  - After you have team members capable of doing your job, you need to give them opportunities to take on more responsibilities or occasionally lead the team
- Know when to make waves
  - Waiting only delaying the inevitable and causing untold damage in the process. So, act and act quickly
- Shield your team from chaos
  - Outside your team is a world of chaos and uncertainty. Chaos has always been present, but manager must shield the team from it.
- Give your team air cover
  - Share as much information as possible with the team, but don't distract team with organizational matters that is extremely unlikely to ever actually affect team
- Let your team know when they're doing well
  - Provide positive feedback often enough
- It's easy to say "yes" to something that's easy to undo
  - Good leaders have a good sense for when something can be undone, but more things are undoable than you think (and this applies to both technical and nontechnical decisions).

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

SCALE INTO A GOOD LEADER

"The three Always of leadership": Always Be **Deciding**, Always Be **Leaving**, Always Be **Scaling**

- Always Be Deciding
  - Managing a team of teams means making ever more decisions at ever-higher levels
  - High-level strategy rather than how to solve any specific engineering task
  - Finding the correct set of trade-offs.
- Always Be Leaving
  - Get the organization to solve problems by itself
  - Build a "Self-Driving" Team
- Always Be Scaling
  - Most precious resource is limited pool of time, attention, and energy
  - Work mostly on important things, rather than urgent things

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PRODUCT

- Examine the product and the problem it is intended to solve at the very beginning of the project
- Software Scope → unambiguous and understandable at the management and technical levels
  - Context → How does the software to be built fit into a larger system, product, or business context, and what constraints are imposed as a result of the context?
  - Information objectives → What customer-visible data objects are produced as output from the software? What data objects are required for input?
  - Function and performance → What function does the software perform to transform input data into output? Are any special performance characteristics to be addressed?

PRODUCT: ENGINEERING FOR EQUITY

- Software engineering → solve problems
- Not only solve problems, build products that drive **positive outcomes** for the broadest base of people, including people who are the most difficult to reach
- Consider how the tools you build will be leveraged to change the trajectory of humanity

PRODUCT: BIAS IS THE DEFAULT

- Fail to focus on users of different nationalities, ethnicities, races, genders, ages, socioeconomic statuses, abilities, and belief systems → inadvertently can fail users
- Such failures are often unintentional
- People have certain biases → most people exhibit **unconscious bias**, enforcing and promulgating existing stereotypes
- Why is it a important thing to be realized? Look at the next case study

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

CASE STUDY: GOOGLE MISSES THE MARK ON RACIAL INCLUSION

- In 2015, software engineer Jacky Alcine pointed out that the image recognition algorithms in Google Photos were classifying his black friends as “gorillas.” Google was slow to respond to these mistakes and incomplete in addressing them.
- What caused such a monumental failure? Several things:
  - Image recognition algorithms depend on being supplied a “proper” (often meaning “complete”) dataset. The photo data fed into Google’s image recognition algorithm was clearly incomplete. In short, the data did not represent the population.
  - Google itself (and the tech industry in general) did not (and does not) have much black representation, and that affects decisions subjective in the design of such algorithms and the collection of such datasets. The unconscious bias of the organization itself likely led to a more representative product being left on the table.
  - Google’s target market for image recognition did not adequately include such underrepresented groups. Google’s tests did not catch these mistakes; as a result, the users did, which both embarrassed Google and harmed the users.
- As late as 2018, Google still had not adequately addressed the underlying problem

- As a result, the algorithm returned results that caused harm to our users
- The harm to Google itself → reduced user trust and engagement with the company

PRODUCT: UNDERSTANDING THE NEED FOR DIVERSITY

- Exceptional engineer → should bring diverse perspectives into product design and implementation
- Have a computer science degree is insufficient for inclusive and equitable engineering
- Focus all work within the framing of the complete ecosystem to be influenced
  - At least → understand the population demographics of the users

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PRODUCT: BUILDING MULTICULTURAL CAPACITY

- Exceptional engineer → able to understand how products can advantage and disadvantage different groups of human beings
- Beside technical aptitude, it is important to have the discernment to know when to build something and when not to
- Discernment includes building the capacity to identify and reject features or products that drive adverse outcomes

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PRODUCT: MAKING DIVERSITY ACTIONABLE

- Systemic equity and fairness are attainable → willing to accept that:
  - We are accountable for the systemic discrimination we see in the technology sector
  - We are accountable for the failures in the system
- Technology lead or software engineering manager has the means to augment equity on their teams

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PRODUCT: REJECT SINGULAR APPROACHES

- Perpetuate solutions that present a single philosophy or methodology for fixing inequity in the technology sector is not enough
- Problems are complex and multifactorial
- Design from the start and raising development standards for development in order to make software delightful and accessible for people who struggle to access technology can **enhance the experience for all users**
- Best practice → designing for the user who is least like you

PROCESS

- A generic process framework for software engineering encompasses five activities:
  - Communication
  - Planning
  - Modeling
  - Construction
  - Deployment
- Decide which process model is most appropriate for:
  - The customers who have requested the product and the people who will do the work
  - The characteristics of the product
  - The project environment in which the software team works

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PROCESS: KNOWLEDGE SHARING

No matter the format, documentation is important

Building on top of others' work in a community-like way can be an accelerator, both in open source and in companies. Documentation often signals if a repository is reliable to reuse code from, or if it's an active project to contribute to. What signs do developers look for?

In both open source projects and enterprises, developers see about

50%

productivity boost with easy-to-source documentation

What the data shows: At work, developers consider documentation trustworthy when it is up-to-date (e.g., looking at time-stamps) and has a high number of upvotes from others. Open source projects use READMEs, contribution guidelines, and GitHub Issues, to elevate the quality of any project, and to share information that makes them more attractive to new contributors. Enterprises can adopt the same best practices to achieve similar success.

In both environments, developers see about a 50% productivity boost when documentation is up-to-date, detailed, reliable, and comes in different formats (e.g. articles, videos, forums).

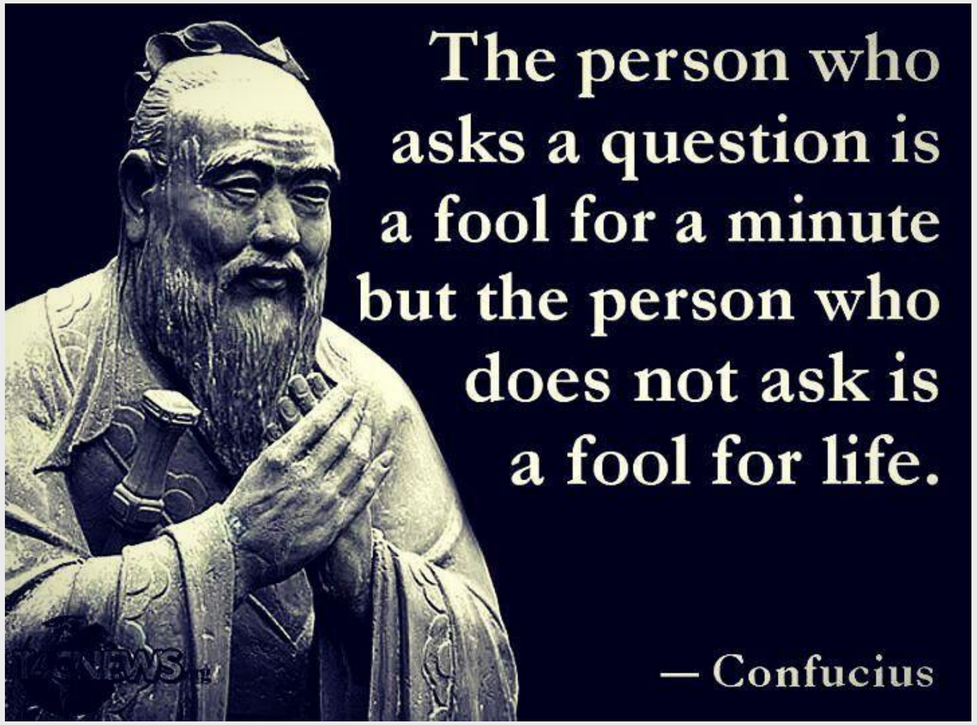
Using the data: Review the documentation your team consumes: When was the last time it was updated? Can everyone on your team improve the documentation? Check this frequently to stay on track.

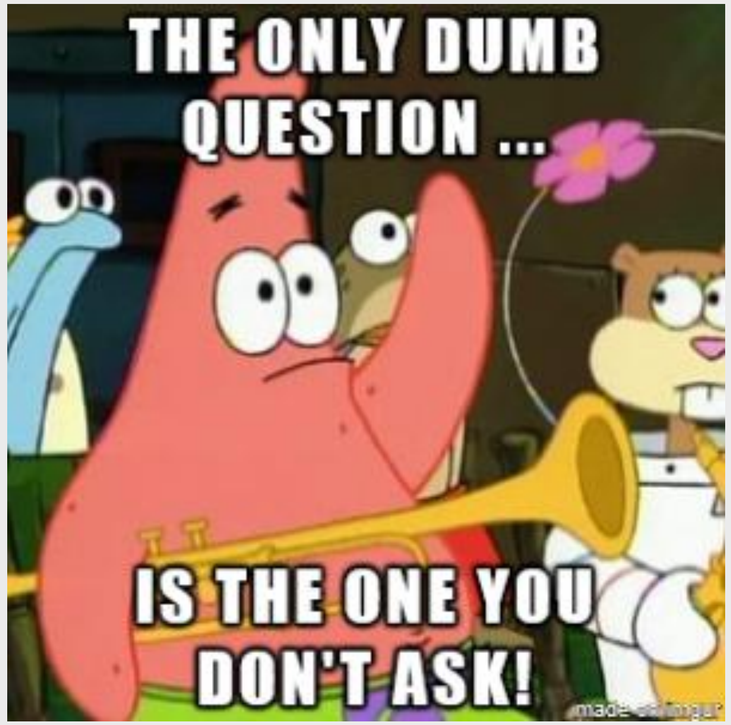
PROCESS: KNOW YOUR AUDIENCE

- Internal document for your team (e.g., meeting note)
- Documentation for project contributors
- Documentation for non-developer collaborators (e.g., UX researchers)
- Documentation for developer users
- Documentation for clients with no software knowledge
- User manual for end users

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PROCESS: ASK THE QUESTIONS





INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PROCESS: WHAT IS WRONG WITH THIS QUESTION?

New To Coding. Can anyone assist me?

Asked 7 years, 1 month ago Modified 7 years, 1 month ago Viewed 47 times

-4

I am trying to make a word counter and I just cant seem to get it. Can anyone help?

```
import re
print("Welcome To This Software Made By Aaron!")
word = raw_input("Enter Your Words: ")
Check = 0
Right = 0
Length = len(word)
while True:
    if Right == 1:
        if Length < Check:
            Check = Check + 1
            print(Check)
    if Length == Check:
        Right = 1

print("Your Word Count Is " +Check)
```

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PROCESS: MAKE IT EASY FOR PEOPLE TO HELP YOU

- I am trying to ____, so that I can ____.
- I am running into ____.
- I have looked at ____ and tried ____.
- + I'm using this tech stack: ____.
- + I'm getting this error/result: ____.
- + I think the problem could be ____.

Avoid Duplication!

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PROCESS: COMMUNICATION AND CONFLICT RESOLUTION

- You can’t solve any problem without communication!

Conflict Resolution

- Your goal: Find a solution to the problem and move forward.
  - As a smart person on “Ted Lasso” once said, “Fight forward, not back.”
- Make sure everybody works from the same set of facts.
- Establish ground rules for your team’s discussion.
  - Talk about how the situation made you feel. Never presume anything about anyone else.
- Remain calm and rational. If you feel triggered or threatened, extract yourself from the situation, wait an hour to chill out, and then try again.
- If you reach an impasse, talk to your team leader.
- If your team remains in conflict, escalate to your mentor.
  - Your mentor will not solve your problem. They will help you to solve your own problems.

INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PROJECT

- To manage a successful software project → understand what can go wrong so that problems can be avoided
  - Understand customer's needs
  - Changes are managed well, includes chosen technology change or shifted business needs and sponsorship management
  - Realistic deadlines
  - End users are ready to accept new system
  - Have the requisite skills
  - Learn from mistakes

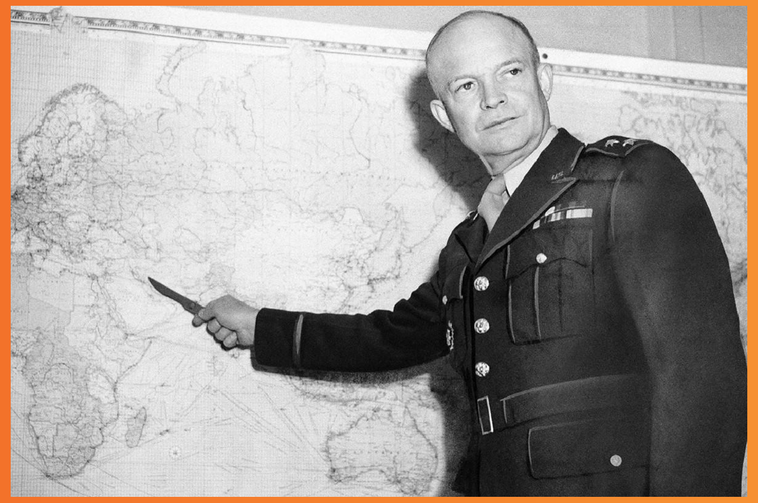
INTRODUCTION TO PROJECT MANAGEMENT CONCEPTS

PROJECT: THE W5HH PRINCIPLE

- An approach that addresses project objectives, milestones and schedules, responsibilities, management and technical approaches, and required resources
  - Why is the system being developed? Assess the validity of business reasons for the software work
  - What will be done? Task set required for the project is defined
  - When will it be done? Establish a project schedule by identifying when project tasks are to be conducted and when milestones are to be reached
  - Who is responsible for a function? The role and responsibility of each member of the software team is defined
  - Where are they located organizationally? Not only software practitioners, the customer, users, and other stakeholders also have responsibilities.
  - How will the job be done technically and managerially? Once product scope is established, a management and technical strategy for the project must be defined.
  - How much of each resource is needed? Developing estimates based on answers to earlier questions

Plans are nothing, planning is everything

- Dwight D. Eisenhower



IMPORTANCE OF PROJECT PLANNING IN SOFTWARE ENGINEERING

SOFTWARE PROJECT PLANNING

- The overall goal of project planning is to establish a pragmatic strategy for controlling, tracking, and monitoring a complex technical project.

- Why?
- So the end result gets done on time, with quality!

IMPORTANCE OF PROJECT PLANNING IN SOFTWARE ENGINEERING

PROJECT PLANNING

Objectives:

- Define Scope and Objectives: Clearly outline what needs to be achieved.
- Resource Allocation: Determine necessary resources.
- Risk Management: Identify potential risks and mitigation strategies.
- Communication Plan: Establish how information will be shared.
- Schedule Development: Create timelines and milestones.

Consequences of Poor Planning

- Cost Overruns
- Missed Deadlines
- Scope Creep
- Reduced Quality
- Team Burnout

WORK BREAKDOWN STRUCTURE (WBS)

DEFINITION AND PURPOSE

- WBS is a hierarchical decomposition of the total scope of work to be carried out by the project team.

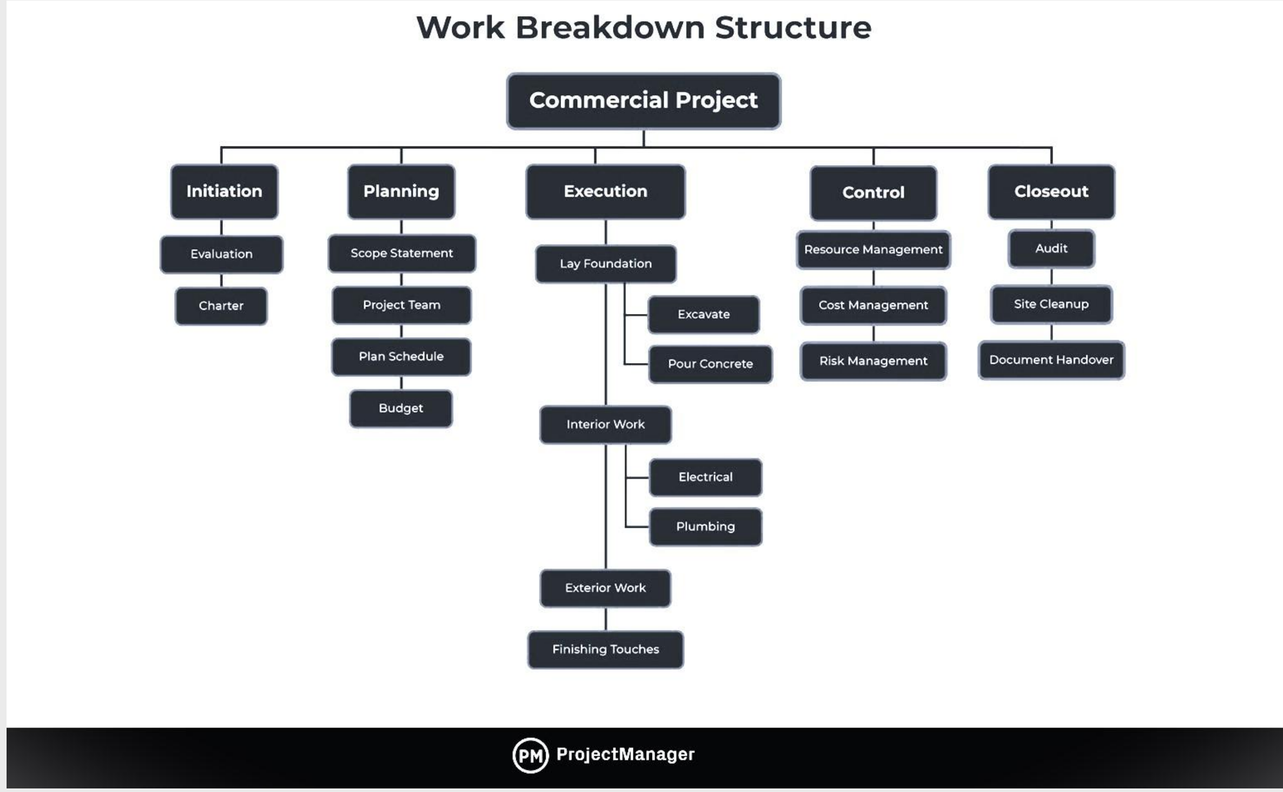
Purpose:

- Organize work into manageable sections.
- Facilitate resource allocation, assignment of responsibilities, and measurement of progress.

WORK BREAKDOWN STRUCTURE (WBS)

STEPS TO CREATE A WBS

- Identify Major Deliverables: Start with the end products or services.
- Decompose Deliverables: Break down into smaller, manageable components.
- Assign Identification Codes: For tracking purposes.
- Verify Decomposition: Ensure completeness and correctness.



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